TECH BIZ SECURITY · SOC ANALYST INTERNSHIP · BATCH B-503

WEEK 2 · TASK 5

Windows Event Logs for SOC Analysts

Submitted by	**Oscar ALIDJINOU**
Student ID	**TBS-INT-2026-2398**
Program	**SOC Analyst Internship (6 weeks)**
Company	**Tech Biz Security**
Date of submission	**27-10-2026**

01 Objective

Windows logging is the core of SOC work: most attacks on a company's computers leave a trail in the Windows event logs. This task had three aims: to learn my way around Event Viewer and the Security log, to build a reference table of the key Event IDs, and to analyse one failed-logon event (Event ID 4625). I also completed Windows Fundamentals 2 and began Investigating Windows to understand the system tools an analyst uses on a Windows host.

02 Proof of Completion

Room	Duration	Tasks	Status
TryHackMe: Windows Fundamentals 1	30 min	10	Completed earlier (see my Week 1 Task 3 report)
TryHackMe: Windows Fundamentals 2	30 min	9	Completed, room progress 100%
TryHackMe: Investigating Windows	45 min	1	Completed, room progress 100%

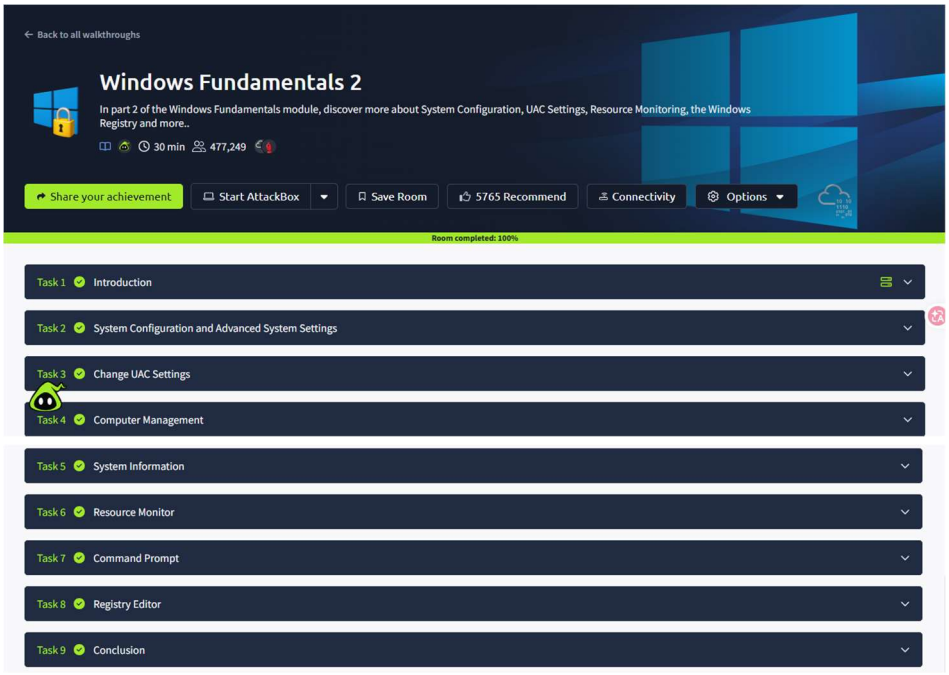


Figure 1. Windows Fundamentals 2: room completed (100%).

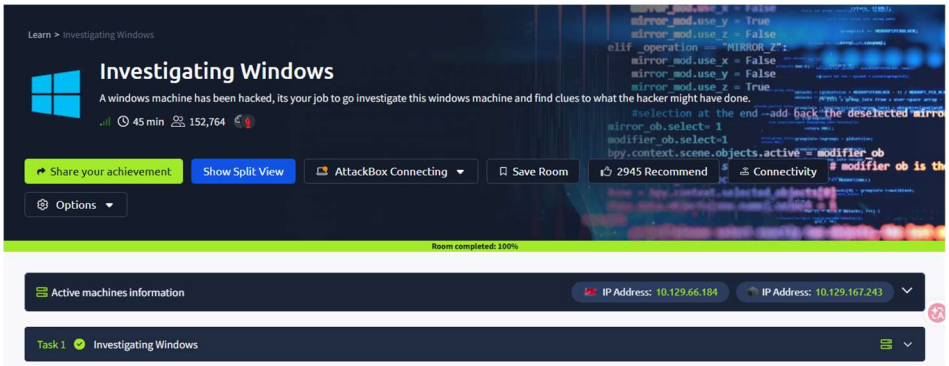


Figure 2. Investigating Windows: room completed (100%).

> **Where I worked**
> The Event Viewer overview and the system tools in this report were run on the Windows Server machine provided by TryHackMe (host THM-WINFUN2). The 4624 and 4625 events were examined on my own Windows computer (host DESKTOP-F5LFQUD), because I could not run my local Windows virtual machine at that moment.

03 Windows Event Viewer

3.1 What it is and how it is laid out

Event Viewer shows the events recorded on a computer, which makes it an audit trail of what the system and its users have done. It has three panes: a tree of log providers on the left, the list and details of the selected events in the middle, and an actions pane on the right. I opened the Security log, which on the lab machine held over 100,000 events.

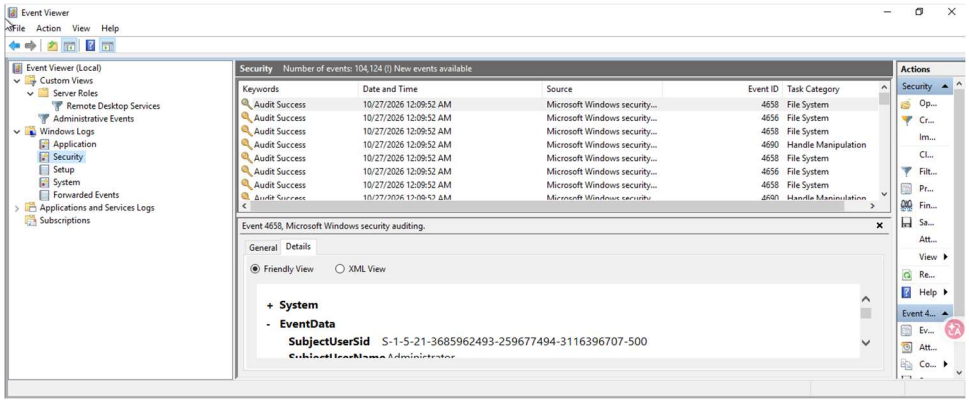


Figure 3. Event Viewer with the Security log selected and one event's details open.

Windows log	What it records
Application	Events from programs, such as an application error or crash.
Security	Audit events: logons, failed logons, account changes, privilege use. The most important log for a SOC analyst.
System	Events from the operating system and its components, such as a driver failing or a service starting.
Setup / Forwarded Events	Installation events, and events collected from other computers.

Each event is classed as Information, Warning or Error, or in the Security log as an audited Success or Failure. Each row shows its keywords, date and time, source, Event ID and task category, and the Details tab shows the full record in a friendly or XML view.

3.2 Finding Event IDs 4624 and 4625

In the Security log I looked at the Event ID column. Successful logons (4624) appear all the time, together with their related events, such as special-privilege logons (4672), logoffs (4634) and account lookups (4798 and 4799). Failed logons (4625) are much rarer, and when they appear they stand out.

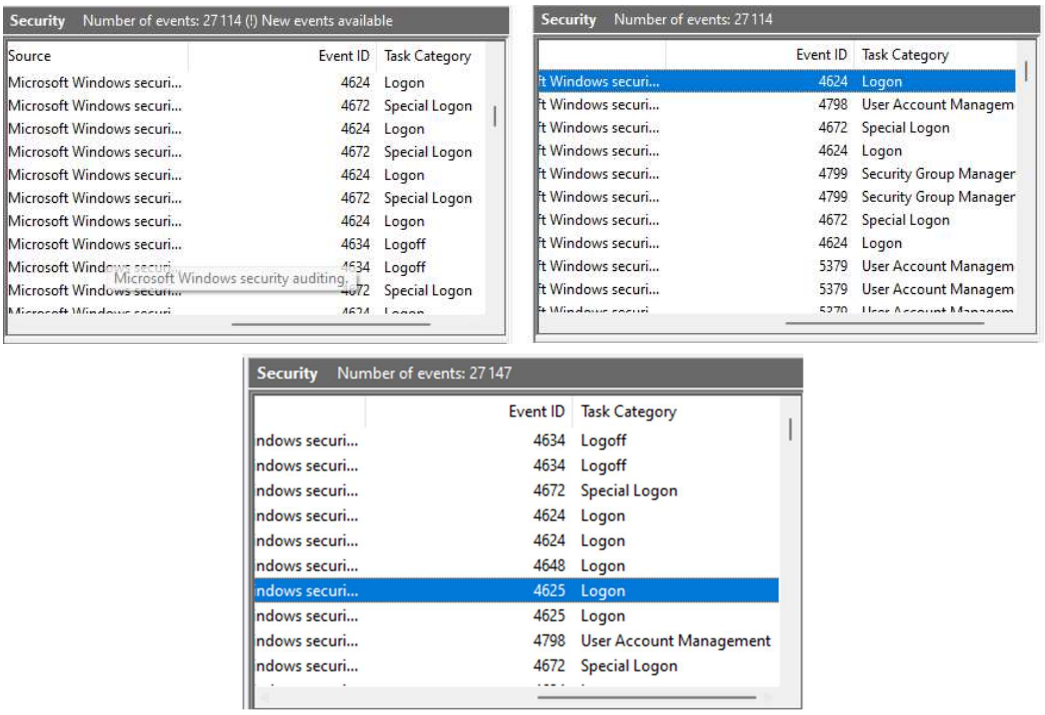


Figure 4. Security log on my Windows computer: logon events (4624), related events and a failed logon (4625, highlighted in the last view).

> **Tip for working with a large log**
> Use Filter Current Log in the Actions pane and type an Event ID such as 4625. This hides everything else and leaves only the events I am hunting for.

04 Five Key Event IDs

ID	Meaning	Why a SOC analyst cares	In my screenshots
4624	An account was successfully logged on	Shows who logged in, how and from where. A success after many failures can mean an account was guessed.	Yes
4625	An account failed to log on	Reveals wrong passwords, brute force and password spraying.	Yes
4688	A new process was created	Records programs that were started, and the parent process. With command-line auditing enabled it shows exactly what was run.	No (needs auditing enabled)
4720	A user account was created	Attackers add accounts to keep access, so an unexpected 4720 is a strong warning sign.	No
4672	Special privileges assigned to a new logon	Marks a logon with administrator-level rights. Normal for admins, suspicious for ordinary users.	Yes

More IDs worth knowing

ID	Meaning	Seen or used for
4634	An account was logged off	Session end; used with 4624 to measure how long a session lasted. Seen in my log.
4648	A logon was attempted using explicit credentials	Credentials typed for another account, as with runas or remote access. Seen in my log.
4798 / 4799	A user's or group's local group membership was enumerated	Looking up who belongs to which group; attackers do this during reconnaissance. Seen in my log.
4732	A member was added to a security-enabled local group	Adding an account to Administrators is a classic privilege step.
4698	A scheduled task was created	Scheduled tasks are a common persistence method.
7045	A new service was installed (System log)	Malicious services used for persistence or remote execution.
1102	The audit log was cleared	Attackers clear logs to hide their tracks; always investigate.

05 Analysis of a Failed-Logon Event (4625)

5.1 The event I selected

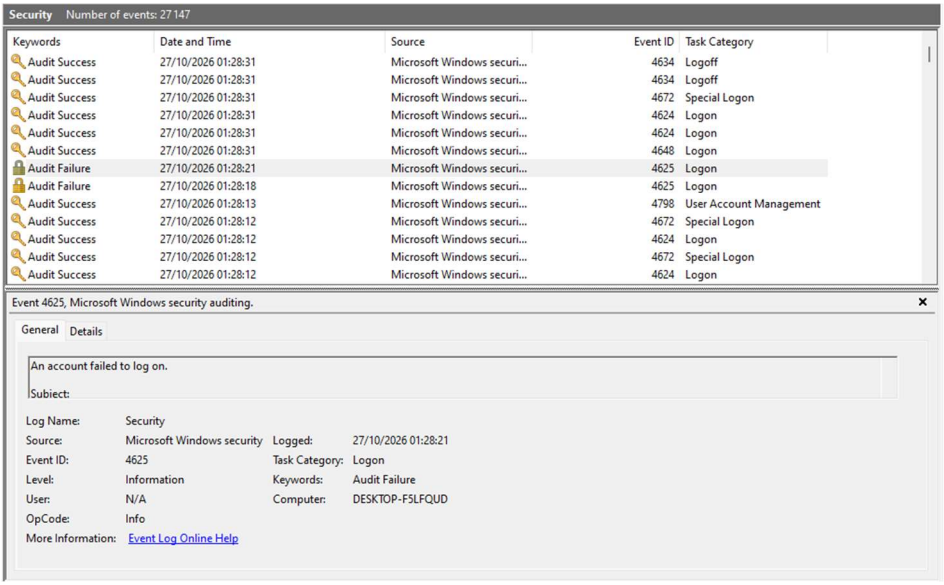


Figure 5. Event 4625 selected in the Security log, with the General tab open.

Field	Value in my event
Log name / source	Security / Microsoft Windows security auditing
Event ID and category	4625, Logon
Keywords	Audit Failure
Logged	27/10/2026 at 01:28:21
Computer	DESKTOP-F5LFQUD
Description	"An account failed to log on."

5.2 The full details of the event

The Details tab of the event shows every field the system recorded. I read them in four groups: who reported the failure, which account was targeted, why it failed, and where the attempt came from.

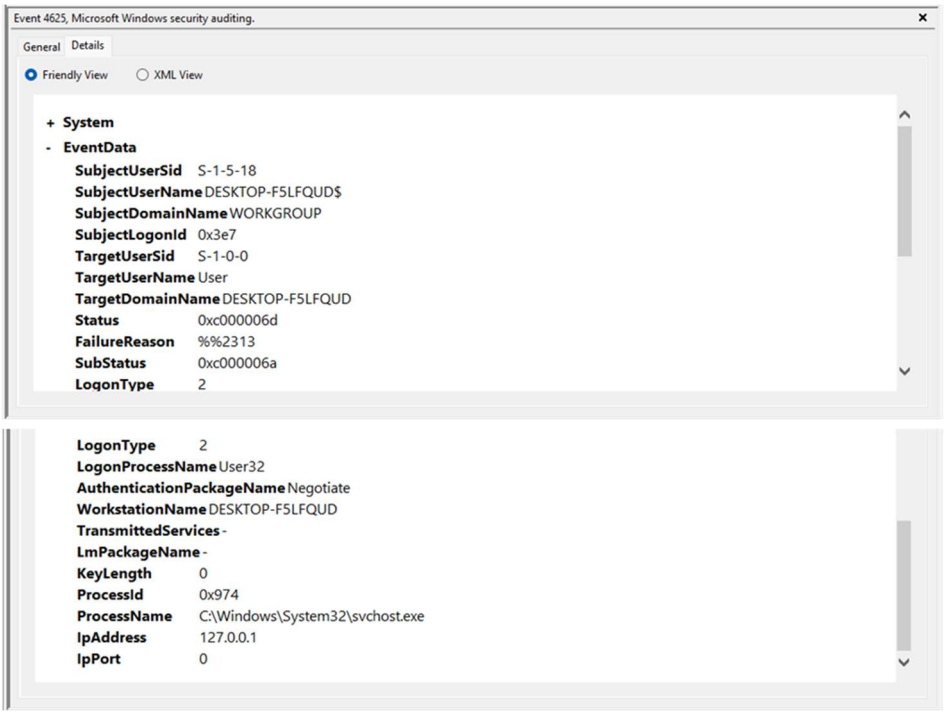


Figure 6. Details tab of the 4625 event (Friendly View), top and bottom of the EventData.

Field	Value	What it tells me
SubjectUserName	DESKTOP-F5LFQUD$ (S-1-5-18)	The computer's own account (Local System) reported the failure, which is normal for a sign-in on this machine.
SubjectDomainName	WORKGROUP	The computer is not joined to a company domain.
TargetUserName	User	The account someone tried to log in to.
TargetDomainName	DESKTOP-F5LFQUD	A local account on this computer, not a domain account.
Status	0xC000006D	Generic logon failure: bad user name or password.
FailureReason	%%2313	Windows' code for "unknown user name or bad password".
SubStatus	0xC000006A	The user name exists but the password is wrong.
LogonType	2 (Interactive)	The attempt was made at the computer itself, not over the network.
LogonProcessName	User32	The Windows sign-in component.
AuthenticationPackage	Negotiate	The standard Windows method of choosing the authentication protocol.
WorkstationName	DESKTOP-F5LFQUD	The attempt came from this same computer.
ProcessName / ProcessId	C:\Windows\System32\svchost.exe / 0x974	A genuine Windows process running from its normal folder (0x974 is 2420 in decimal).
IpAddress / IpPort	127.0.0.1 / 0	The loopback address: the source is the machine itself, with no outside origin.

5.3 What happened around it

An event only makes sense in context, so I read the events before and after it, in time order:

Time	Event IDs	Reading
01:28:12	4624, 4672	Successful logons, including a logon with special privileges.
01:28:13	4798	The user's local group membership was looked up.
01:28:18	4625	First failed logon.
01:28:21	4625	Second failed logon, three seconds later (the event I analysed).
01:28:31	4648, 4624, 4672, 4634	A logon using explicit credentials, then successful logons and logoffs.

5.4 My assessment

Question	Answer from the evidence
What happened?	A password was typed wrongly for the local account "User". Windows refused the logon and recorded 4625.
Where did it come from?	From the computer itself: logon type 2, loopback address 127.0.0.1, same workstation name. No network source.
Is the process legitimate?	Yes. svchost.exe ran from C:\Windows\System32, the normal location.
Was an account guessed?	No. Two failures three seconds apart were followed by successful logons within about ten seconds.
Verdict	Benign. A mistyped password, so a false positive for attack detection. Severity: low.
Action	Close with a note. No escalation or blocking is needed.

> **What would change my verdict**
> Many failures per minute for the same account, a logon type of 3 or 10 (network or Remote Desktop), a source IP other than 127.0.0.1, a sub status of 0xC0000064 (user does not exist) across many names, or a 4624 success right after a long run of failures. Any of these would match brute force (MITRE ATT&CK T1110) and I would escalate.

5.5 How I read any 4625 event

Section of the event	What to check
Subject	The account or process that reported the failure (often the system itself).
Logon Type	How the logon was attempted (see the table below).
Account For Which Logon Failed	The username that was tried. A name that does not exist suggests guessing.
Failure Information (Status and Sub Status)	The exact reason; the codes are in the table below.
Process Information	The program that made the attempt.
Network Information	The workstation name, source IP address and port. This is what I check against threat intelligence.

Logon type	Name	Meaning
2	Interactive	Someone typed the password at the keyboard (my event).
3	Network	Access to a shared folder or other network resource.
4	Batch	A scheduled task.
5	Service	A Windows service starting under an account.
7	Unlock	Unlocking a locked workstation.
10	RemoteInteractive	Remote Desktop. Failures from the internet are a common attack.
11	CachedInteractive	A logon with cached credentials, often on a laptop away from the domain.

Status / Sub Status	Meaning
0xC000006D	General logon failure (bad username or password). Status of my event.
0xC000006A	The username exists but the password is wrong. Sub status of my event.
0xC0000064	The username does not exist.
0xC0000234	The account is locked out.
0xC0000072	The account is disabled.
0xC0000071	The password has expired.
0xC000006F	Logon attempted outside the allowed hours.

5.6 Decision guide

Pattern	Likely meaning and action
One or two failures, then a success	A typing mistake. Close with a note (my case).
Many failures for one account	Password guessing. Check the source and consider a lockout.
Failures across many accounts from one source	Password spraying or enumeration. Escalate.
Logon type 3 or 10 from an external address	Remote attack on a service. Block the source and investigate.
A run of failures followed by a 4624 success	Possible successful compromise. Escalate immediately.

How I would pull them: in Event Viewer, filter the Security log on 4625. In PowerShell, run the command below (administrator rights are needed).

```
Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4625} -MaxEvents 10 | Format-List TimeCreated, Message
```

06 Windows System Tools (Windows Fundamentals 2)

This room covered the tools an analyst uses to look inside a Windows machine. I worked through each on the TryHackMe Windows Server machine.

6.1 System Configuration (msconfig)

MSConfig is a troubleshooting tool for startup problems. It has five tabs: General (normal, diagnostic or selective startup), Boot (boot options and the installed operating system), Services (every configured service, running or not), Startup (programs that start at logon) and Tools (shortcuts to other utilities). On this machine the Startup tab pointed to Task Manager, so I looked at the Startup folder directly.

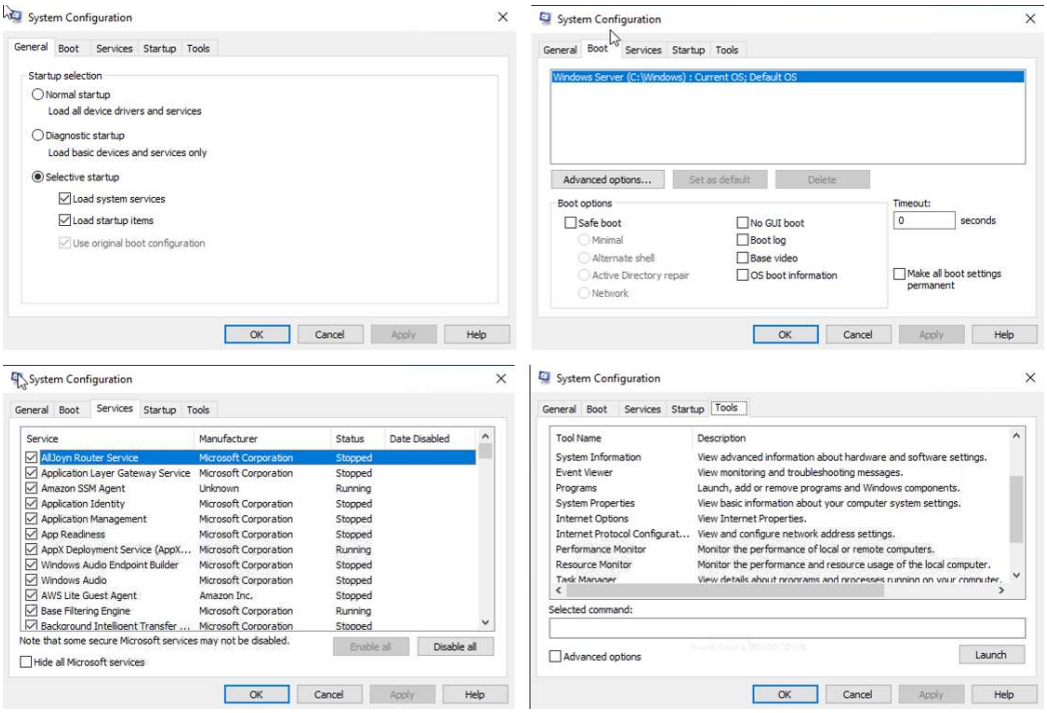


Figure 7. The General, Boot, Services and Tools tabs of System Configuration.

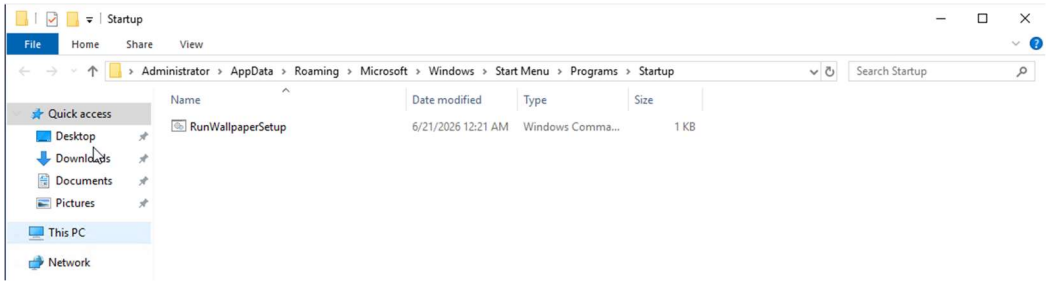


Figure 8. The user Startup folder holds a script that runs at every logon.

> **Security link**
> Services and startup entries are two of the most common places for malware to hide so that it survives a reboot. Checking them is a standard step in any host investigation.

6.2 Advanced System Settings and UAC

Advanced System Settings controls performance (visual effects, processor scheduling, virtual memory) and what happens when the system fails. The paging file here was 704 MB. Startup and Recovery was set to restart automatically and write an automatic memory dump to MEMORY.DMP.

A memory dump is a useful forensic artefact, because it captures what was in memory at the moment of a crash.

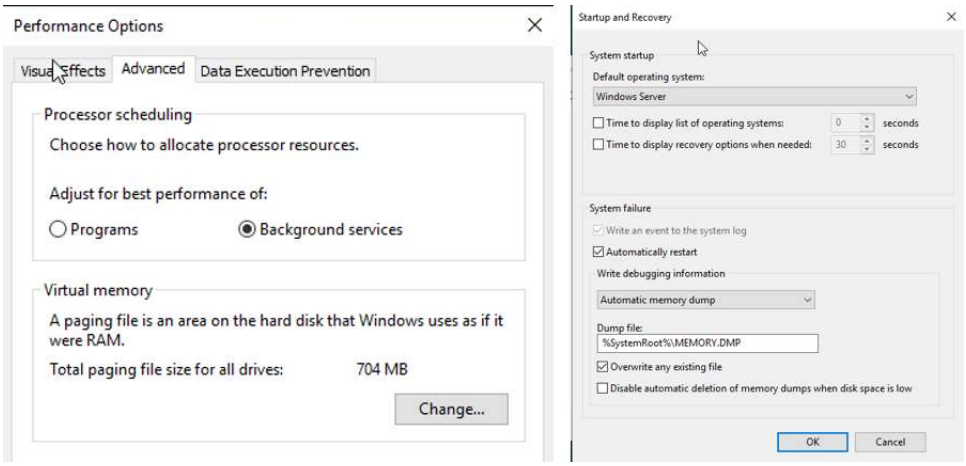


Figure 9. Performance options and the Startup and Recovery settings.

User Account Control has four levels: Always notify (highest security, with the secure desktop dimmed), Notify only when apps try to make changes (the default), the same without dimming, and Never notify. The default level was selected on this machine.

6.3 Computer Management

Computer Management groups System Tools (Task Scheduler, Event Viewer, Shared Folders, Local Users and Groups, Performance), Storage (backup and Disk Management) and Services and Applications. I created a scheduled task named Test that starts a program every day at 8:11 AM, and I saw the default administrative shares ADMIN$, C$ and IPC$ in Shared Folders. Performance Monitor (perfmon) shows performance data live or from a log.

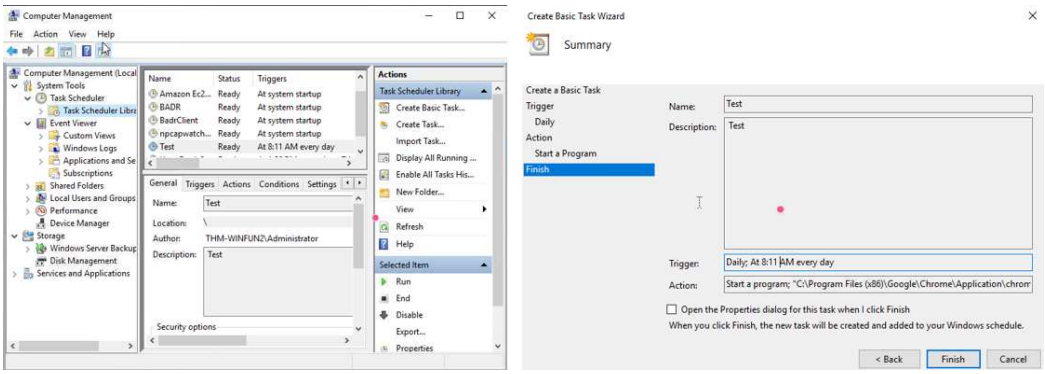


Figure 10. The Task Scheduler library with my new task, and the wizard summary.

> **Security link**
> Scheduled tasks are a favourite persistence mechanism. Creating one is logged as Event ID 4698, so an unexpected task is something I can hunt for.

6.4 System Information and Resource Monitor

System Information (msinfo32) gives one view of the hardware resources, components and software environment of the computer, which helps diagnose problems and understand the host. Resource Monitor (resmon) shows CPU, disk, network and memory use per process, and lets me see which processes are using the network.

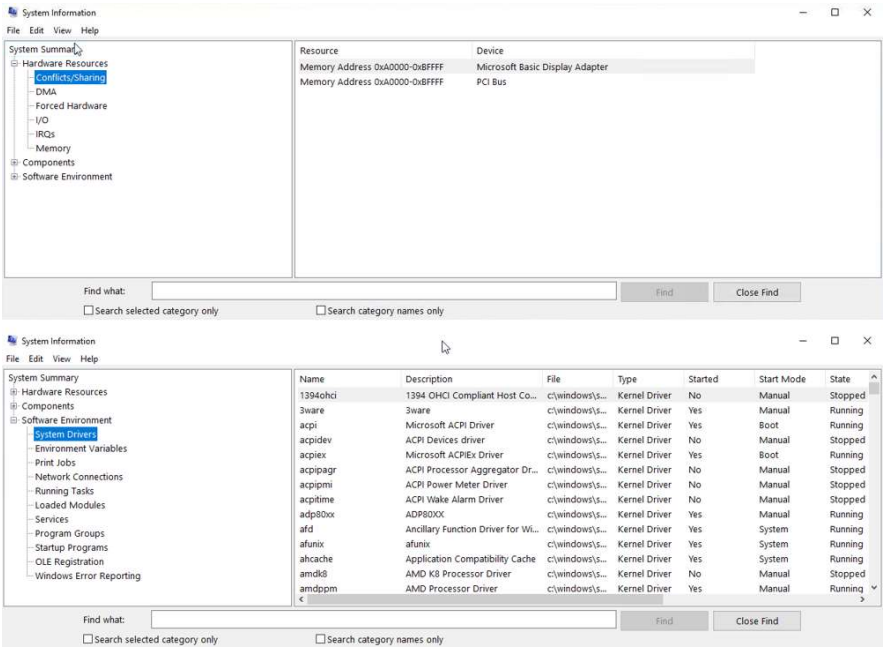


Figure 11. System Information: hardware resources and software environment.

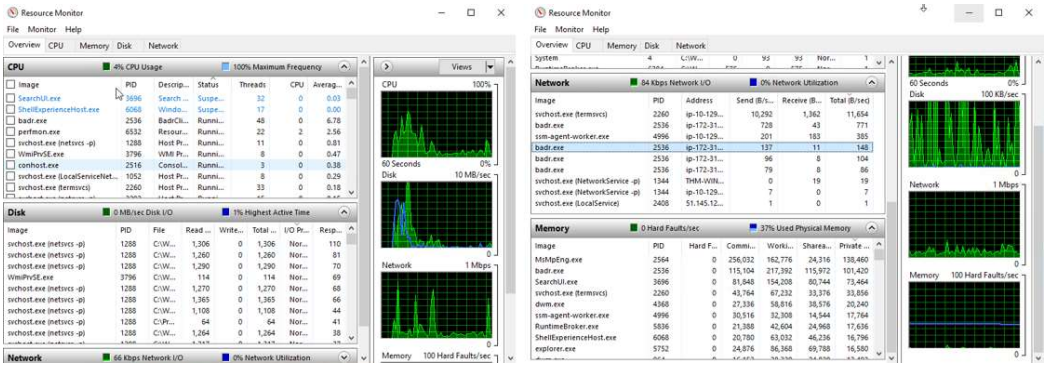


Figure 12. Resource Monitor: CPU, disk, network and memory activity per process.

6.5 Command Prompt

Command	What it shows	Security use
`hostname` `whoami`	The computer name and the current user.	Confirms which machine and account I am working as.
`ipconfig /all`	Full network configuration: IP address, gateway, DHCP and DNS servers.	Identifies the host on the network.
`netstat`	Active connections and their states.	Spots unexpected connections, such as an unknown remote address.
`net user`	The local user accounts.	Reveals accounts that should not exist.
`net accounts`	The password and lockout policy.	Shows how well the system resists password guessing.

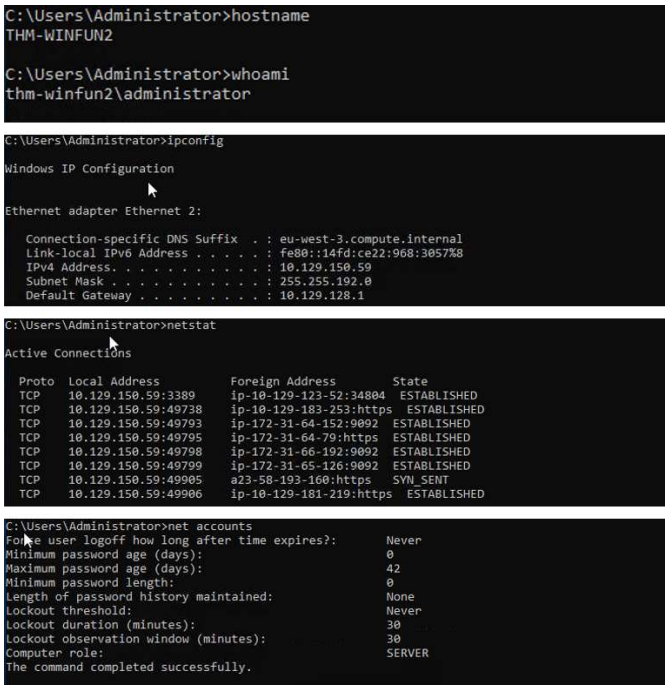


Figure 13. hostname and whoami, ipconfig, netstat and net accounts on the lab machine.

> **A weakness I noticed**
> net accounts showed a minimum password length of 0 and a lockout threshold of "Never". With no lockout, nothing stops an attacker from guessing passwords endlessly, which would produce exactly the stream of 4625 events I analysed in section 05.

6.6 Registry Editor

The Windows Registry is the central database of configuration for the system, users, applications and hardware. Its five root keys are HKEY_CLASSES_ROOT, HKEY_CURRENT_USER, HKEY_LOCAL_MACHINE, HKEY_USERS and HKEY_CURRENT_CONFIG. Changes can affect how Windows runs, so I only browsed it.



Figure 14. The five root keys of the Registry.

> **Security link**
> The Run keys under HKEY_LOCAL_MACHINE and HKEY_CURRENT_USER, in Software\Microsoft\Windows\CurrentVersion, start programs at logon, so attackers use them for persistence.

07 Investigating Windows

This room puts a compromised Windows machine in front of me and asks me to find what the attacker did. The first step in any investigation is to know exactly what system I am looking at, so I used systeminfo, filtered with findstr to show only the lines I needed.

```
systeminfo | findstr /B /C:"OS Name" /C:"OS Version"
```

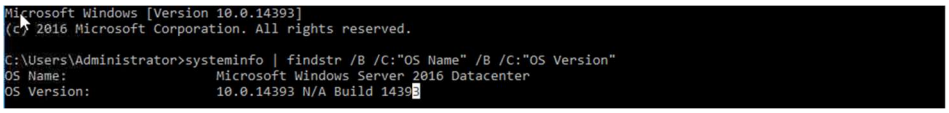


Figure 15. Identifying the operating system with systeminfo and findstr.

The result identified the machine as Windows Server 2016 Datacenter, build 10.0.14393. Knowing the version and build tells me which vulnerabilities and which log behaviour to expect.

A checklist for a suspect Windows host

- **Accounts:** net user and Local Users and Groups, looking for new or unexpected accounts.
- **Persistence:** scheduled tasks, services, startup folders and Registry Run keys.
- **Connections:** netstat for unusual remote addresses.
- **Processes:** Task Manager and Resource Monitor for names, paths and network use.
- **Logs:** Event Viewer for 4624, 4625, 4720, 4732, 4698, 7045 and 1102.

08 The Defender's View

Windows area	How attackers abuse it	What I look for
Logons	Guessing passwords against local or remote accounts.	Bursts of 4625, then a 4624 success; logon types 3 and 10.
Accounts	Creating a hidden or backup account.	Event 4720, and group changes (4732).
Scheduled tasks	Running a payload at every logon or at a set time.	Event 4698 and unfamiliar tasks in Task Scheduler.
Services and startup	Starting malware at boot.	Event 7045 and unknown entries in startup locations.
Registry Run keys	Starting a program at logon.	Unexpected values in the Run keys.
Audit logs	Clearing events to cover tracks.	Event 1102.

09 Reflection and Key Takeaways

- The Security log is the first place to look for account activity, and a few Event IDs (4624, 4625, 4672, 4688, 4720) answer most early questions.
- A single failed logon means little. Its details (logon type 2, a sub status of wrong password, a loopback source) and the events around it showed that mine was a typing mistake, not an attack.
- Every Windows tool I learned this week, from msconfig to the Registry, doubles as a place to look for persistence.
- Weak settings such as no lockout policy make the logs I analyse more dangerous, so detection and hardening go together.

Next step: generate and compare several failed-logon patterns in my own lab, and practise reading 4688 events with command-line auditing enabled.

10 References

- TryHackMe: Windows Fundamentals 1 and 2 (free rooms)
- TryHackMe: Investigating Windows (free room)
- Microsoft documentation: Windows security auditing events (4624, 4625 and related)